Medicine, Dentistry and Medical laboratory

Cell Biology

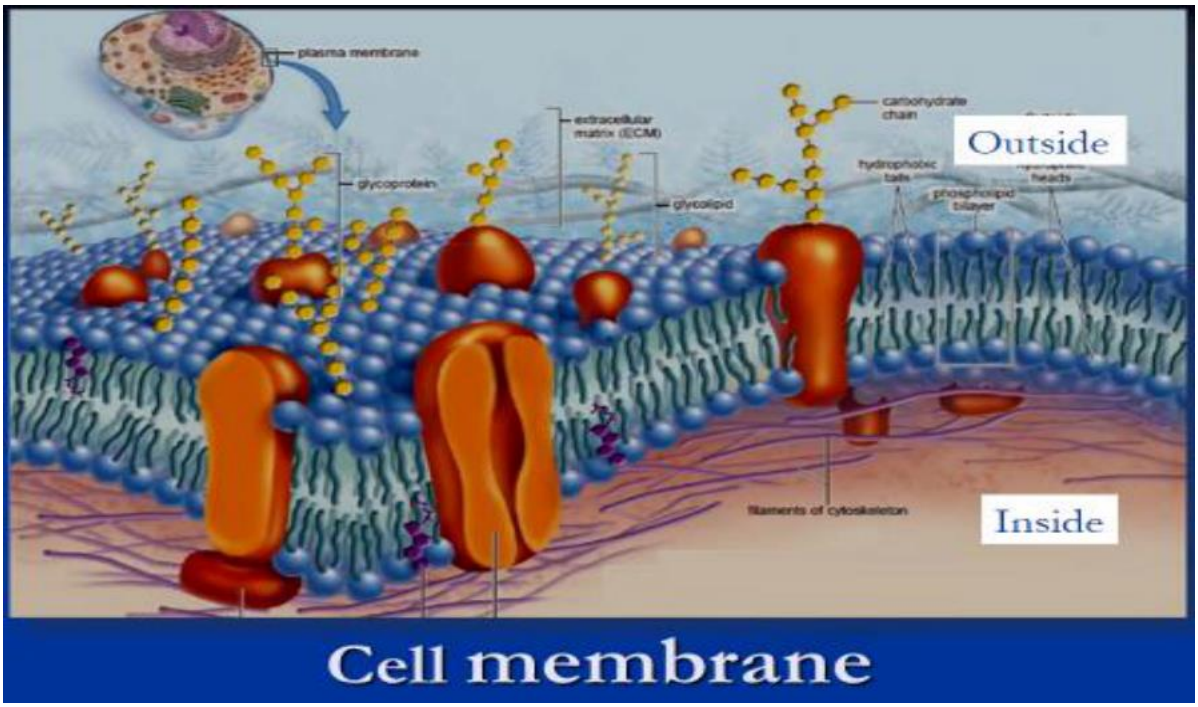
Lecture 5

Cell membrane structure and function

Plasma membrane of an animal cell:

***** The membrane is composed of a **phospholipid bilayer** in which **proteins** are **embedded** (**integral proteins**) or also occur on the cytoplasmic side (**peripheral proteins**). The **hydrophilic heads** of phospholipids are a part of the **outside** surface and the i**nside** surface of the membrane. The **hydrophobic tails** make up the **interior** of the membrane.

***** Note the plasma membrane's asymmetry—**carbohydrate chains** are attached to the **outside** surface and **project** into the **extracellular matrix**. **Cytoskeleton filaments** are attached to the **inside** surface.



Phospholipids bilayer:

***** A phospholipid is an **amphipathic** molecule, meaning that it has both a **hydrophilic (**water-loving**)** region and a **hydrophobic (**water-fearing**)** region.

***** The amphipathic nature of phospholipids largely explains why the membrane is **a bilayer**—has **two** layers of phospholipids.

***The hydrophilic polar heads** of phospholipids are a part of the **outside** surface and the **inside surface** of the membrane.

* **The hydrophobic nonpolar tails** face each other.
* The hydrophobic tails make up the interior of the membrane.

Carbohydrate Chains:

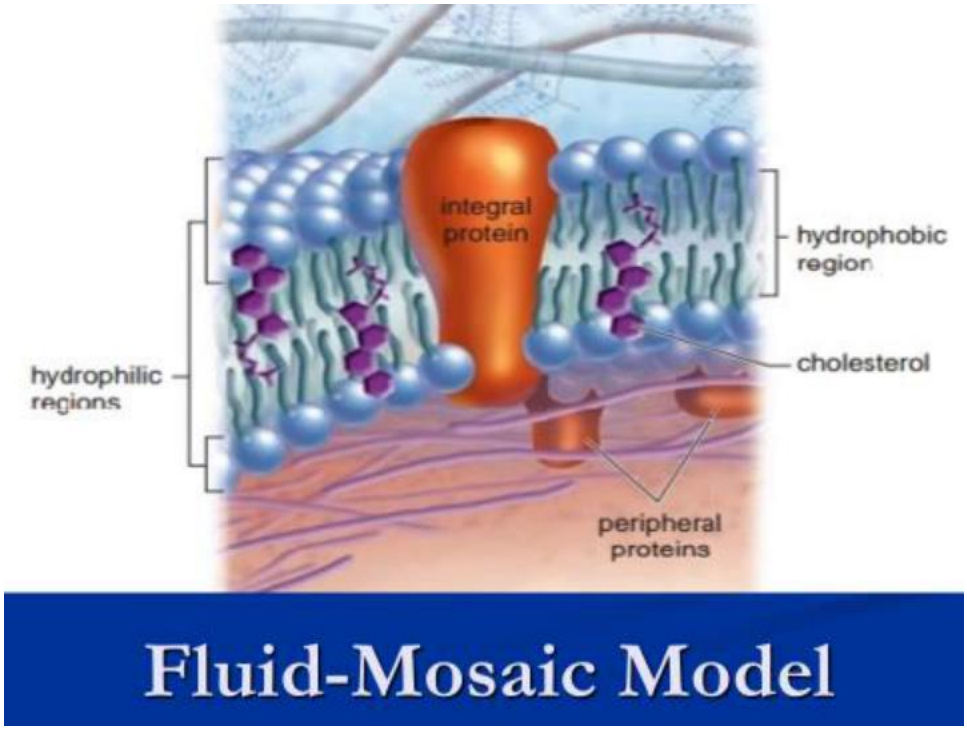
***** The carbohydrate chains are attached to the outside surface and project into the extracellular matrix (**ECM**).

***** Only animal cells have an extracellular matrix (**ECM**), which contains various protein fibers and also very large and complex carbohydrate molecules.

***** The **ECM** has various functions, from lending support to the plasma membrane to assisting communication between cells.

* Both phospholipids and proteins can have attached carbohydrate (sugar) chains.
* These molecules are called **glycolipids** and **glycoproteins**, respectively.
***** In animal cells, the carbohydrate chains of proteins give the cell a "**sugar coat**," more properly called the **glycocalyx**.
***** The glycocalyx protects the cell and has various other functions.

Fluid-Mosaic Model:



The model currently in use to describe the plasma membrane is called the **fluid-mosaic model**.

* The fluid-mosaic model of membrane **structure** refers to the **fluidity** of phospholipids and the **pattern** of proteins in the membrane.

* The fluidity of the membrane is due to its lipid component.

*At **body** temperature, the phospholipid bilayer of the plasma membrane has the consistency of olive oil. The greater the concentration of unsaturated fatty acid residues, the more fluid is the bilayer.

* Cholesterol is another lipid found in the animal plasma membrane.

* The presence of cholesterol molecules in the plasma membrane affects its fluidity.

* At **higher** temperatures, cholesterol stiffens the membrane and makes it less fluid than it would otherwise be.

***** At **lower** temperatures, cholesterol helps prevent the membrane from freezing by not allowing contact between certain phospholipid tails.

* The mosaic **nature** of the plasma membrane is due to its protein content. The number and kinds of proteins can vary in the plasma membrane and in the membrane of the various organelles.

***** The embedded proteins are termed **integral** proteins, and other proteins that occur only on the cytoplasmic side of the membrane are termed **peripheral** proteins.

***** Cytoskeleton filaments are attached to the **inside** surface and fibers of the extracellular matrix to the **outside.**

Permeability of the Plasma Membrane:
The plasma membrane keeps the organelles and the cytoplasm inside the cell and regulates the passage of molecules into and out of the cell.

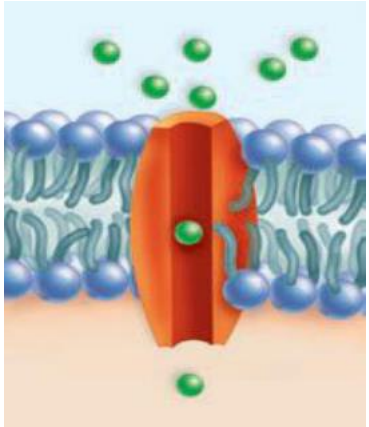
***** This function is critical **because** the life of the cell depends on **maintenance** of its normal composition.

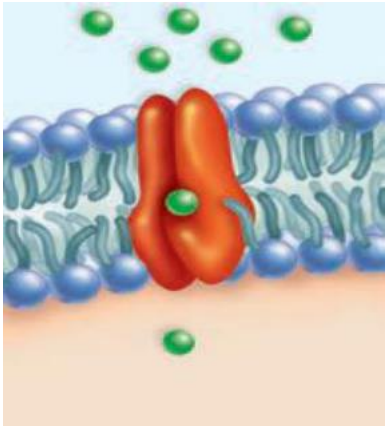
***** The plasma membrane can carry out this function because it is **differentially (selectively) permeable**, meaning that certain substances **can** move across the membrane while others **cannot.**

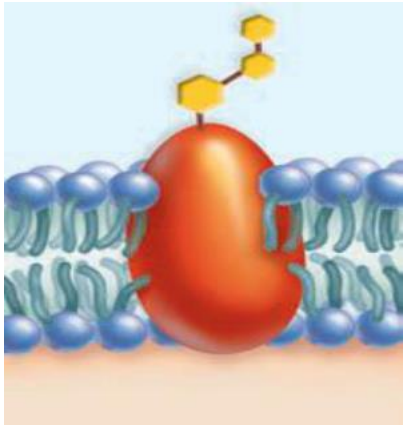
The Functions of the Proteins:

Membrane protein diversity:

1- Channel Protein. **2- Carrier Protein.** **3- Cell recognition. Proteins.**







1- Channel Protein:

*****Allows a particular molecule or ion to cross the plasma membrane freely. They have a channel that allows a substance to simply move across the membrane. For example, a channel protein allows hydrogen ions to flow across the inner mitochondrial membrane. Without this movement of hydrogen ions, ATP would never be produced.

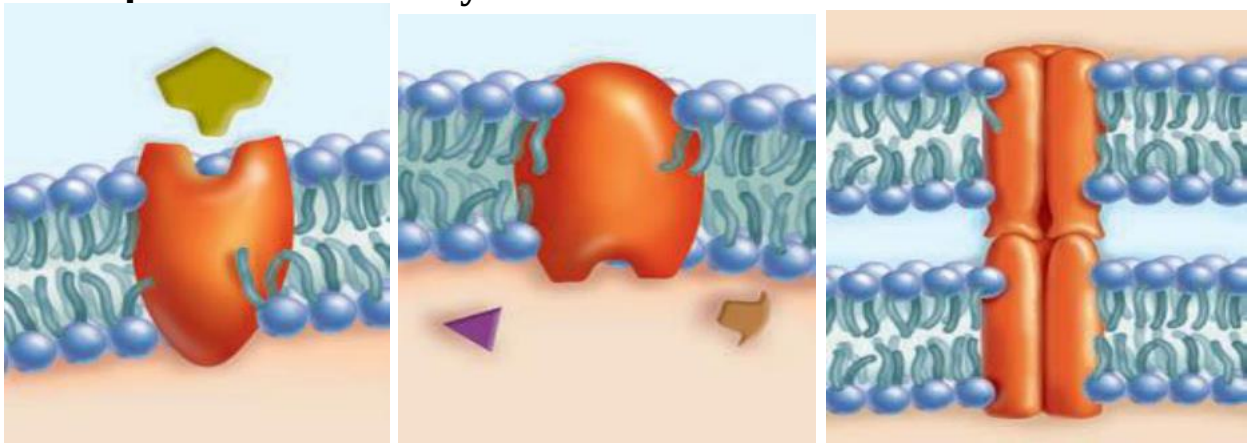
2- Carrier Protein:

*** Selectively** interacts with a **specific** molecule or ion so that it can cross the plasma membrane. They **combine** with a substance and help it move across the membrane. A carrier protein transports **sodium** and **potassium** ions across the plasma membrane of a nerve cell. **Without** this carrier protein, nerve conduction would be impossible.

3- Cell Recognition Protein:

* Cell recognition proteins are **glycoproteins**. Among other functions, these proteins help the body **recognize** when it is being invaded by pathogens so that an immune response can occur. Without this recognition, pathogens would be able to **freely invade** the body. Cells with foreign major histocompatibility complex (MHC) glycoproteins are attacked by **white** blood cells responsible for immunity.

4- Receptor Protein. 5- Enzymatic Protein. 6- Junction Proteins



4- Receptor Protein:

***** Is **shaped** in such a way that a specific molecule can bind to it. The binding of this molecule causes the protein to change its shape and **thereby** bring about a cellular response.

5- Enzymatic Protein:

*** Some** plasma membrane proteins are **enzymatic** proteins that carry out metabolic reactions directly. Without the presence of enzymes, some of which are attached to the various membranes of the cell, a cell would never be able to perform the metabolic reactions necessary to its proper function.

6- Junction Proteins:

Proteins are involved in forming various types of junctions between animal cells.

They are subdivided into:

1- Adhesion junctions.

2-Tight junctions.

3- Gap junctions

***** Tight junctions **join** cells so that a tissue can fulfill a function, as when a tissue pinches off the neural tube during development. Without this cooperation between cells, an animal embryo would have no nervous system.